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BASIC MATHEMATICS

FORM TWO

Form Two National Assessment (FTNA) 2021 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Two
Year	2021
Exam Board	NECTA
Exam	Form Two National Assessment (FTNA)
Type	Answer key and worked solutions
Questions	10 compulsory questions; 28 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F2-2021.pdf

Question 1 — Numbers and approximation

1(a)(i). Write 498,030 in words.

Step 1: Split into thousands and the remainder: 498 thousand + 30.

Step 2: Write each group in words and join them.

Final Answer: Four hundred and ninety-eight thousand and thirty.

1(a)(ii). Express 498,030 in standard notation.

Step 1: Move the decimal point five places left so the first factor is between 1 and 10.

Step 2: The power of ten is positive because the original number is large.

Final Answer: 4.9803×10^5 .

1(a)(iii). Use listing to find the LCM of 3, 10 and 15.

Step 1: Multiples of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. Multiples of 10: 10, 20, 30.

Step 2: Multiples of 15: 15, 30. The first positive number common to all lists is 30.

Final Answer: 30.

1(b)(i). Write nine hundred ninety-nine million nine hundred ninety-nine thousand nine hundred and one in numerals.

Step 1: Separate millions, thousands and units: 999 | 999 | 901.

Step 2: Each group after the first has three digits; combine them with separators.

Final Answer: 999,999,901.

1(b)(ii). Give the significant figures of 400,780 and 0.00606; round each to one significant figure.

Step 1: Zeros between nonzero digits count; zeros before the first nonzero digit do not. In the usual convention for this written integer, the unmarked final zero is not counted.

Step 2: For 400,780 keep 4 and inspect the next digit 0. For 0.00606 keep the first 6 and inspect the next digit 0.

**Final Answer: 400,780: 5 significant figures; to 1 s.f., $400,000 = 4 \times 10^5$.
0.00606: 3 significant figures; to 1 s.f., $0.006 = 6 \times 10^{-3}$.**

Study Tip: Without a precision marker, the trailing zero in an integer can be ambiguous. If measured to the nearest unit, 400,780 would have six significant figures. Its one-significant-figure rounding is unchanged.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 2 — Fractions and percentages

2(a)(i). Arrange $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{8}$ and $\frac{1}{2}$ from smallest to largest.

Step 1: Use the common denominator 24: $\frac{16}{24}$, $\frac{18}{24}$, $\frac{15}{24}$, $\frac{12}{24}$ respectively.

Step 2: Order the numerators 12, 15, 16, 18, then restore the original fractions.

Final Answer: $\frac{1}{2} < \frac{5}{8} < \frac{2}{3} < \frac{3}{4}$.

2(a)(ii). Find the product of $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{8}$ and $\frac{1}{2}$.

Step 1: Multiply the numerators and denominators: $(2 \times 3 \times 5 \times 1)/(3 \times 4 \times 8 \times 2) = 30/192$.

Step 2: Cancel common factors or divide both numerator and denominator by 6 to get $\frac{5}{32}$.

Final Answer: $\frac{5}{32}$.

2(b). Subtract 0.02 of Tsh 270,000 from 36% of Tsh 50,000 (the digit 2 repeats).

Step 1: Let $d = 0.02222\dots$; then $100d - 10d = 2$, so $90d = 2$ and $d = 1/45$.

Step 2: $(1/45) \times 270,000 = 6,000$; $(36/100) \times 50,000 = 18,000$.

Step 3: Subtract in the stated order: $18,000 - 6,000 = 12,000$.

Final Answer: Tsh 12,000.

Study Tip: The dot above 2 means recurring. Reading the number as the terminating decimal 0.02 changes the answer.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 3 — Units and simple interest

3(a). Find 500 cm + 3150 mm + 3.5 m, in metres.

Step 1: Convert: $500 \text{ cm} = 5 \text{ m}$; $3150 \text{ mm} = 3.15 \text{ m}$.

Step 2: Add: $5 + 3.15 + 3.5 = 11.65 \text{ m}$.

Final Answer: 11.65 m.

3(b). For how many years will Tsh 20,000 earn Tsh 4,800 at 4% per annum?

Step 1: Use simple interest $I = Prt/100$.

Step 2: Rearrange: $t = 100I/(Pr) = (100 \times 4,800)/(20,000 \times 4) = 6$.

Final Answer: 6 years.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 4 — Geometry

4(a)(i). Name polygon ABCD in the figure, whose opposite sides are parallel and all four sides have equal marks.

Step 1: Opposite sides are parallel, so it is a parallelogram.

Step 2: All four sides are equal, identifying the special parallelogram as a rhombus.

Final Answer: Rhombus.

4(a)(ii). In the rhombus, $B = 150^\circ$, $A = x$, $C = (2x - 30)^\circ$ and $D = y$. Find x and y .

Step 1: Opposite angles are equal: $x = 2x - 30$, hence $x = 30$.

Step 2: D is opposite B, so $y = 150^\circ$. Check adjacent angles: $30 + 150 = 180^\circ$.

Final Answer: $x = 30^\circ$; $y = 150^\circ$.

4(b). Find the area of the figure made from square OABC and the remaining three quarters of a circle centred at O; $BC = 7$ cm.

Step 1: The square has side 7 cm; OA and OC are radii, so the circle radius is 7 cm.

Step 2: The square replaces the circle's upper-right quarter. Required area = $7^2 + (3/4)\pi \times 7^2$.

Step 3: Area = $49 + (3/4) \times (22/7) \times 49 = 49 + 115.5 = 164.5 \text{ cm}^2$.

Final Answer: 164.5 cm^2 (using $\pi = 22/7$).

Study Tip: Adding the entire circle to the square would count the upper-right quarter twice.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 5 — Algebra

5(a). A father is three times as old as his son; their total age is 64. Find their ages.

Step 1: Let the son be s , so the father is $3s$.

Step 2: $s + 3s = 64$ gives $4s = 64$ and $s = 16$. Then $3s = 48$; $16 + 48 = 64$.

Final Answer: Son: 16 years; father: 48 years.

5(b). Factorise to solve $x^2 + 7x + 12 = 0$.

Step 1: Find two numbers with sum 7 and product 12: 3 and 4.

Step 2: $(x + 3)(x + 4) = 0$; set each factor equal to zero.

Final Answer: $x = -3$ or $x = -4$.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 6 — Coordinates and transformations

6(a). Find the slope and equation through A(6,4) and B(12,6).

Step 1: $m = (6 - 4)/(12 - 6) = 2/6 = 1/3$.

Step 2: Point-slope form: $y - 4 = (1/3)(x - 6)$, which simplifies to $y = x/3 + 2$.

Final Answer: Slope = $1/3$; equation $y = x/3 + 2$ (or $x - 3y + 6 = 0$).

6(b)(i). A translation takes (0,0) to (-3,-4). Find the image of Q(1,-2).

Step 1: The translation vector is $(-3, -4)$.

Step 2: Add it to Q: $(1 - 3, -2 - 4) = (-2, -6)$.

Final Answer: $Q' = (-2, -6)$.

6(b)(ii). Reflect $A(-5,2)$ and $B(4,-7)$ in the y-axis.**Step 1:** Reflection in the y-axis maps (x,y) to $(-x,y)$.**Step 2:** Change only the sign of each x-coordinate.**Final Answer: $A'(5,2)$; $B'(-4,-7)$.**

Section Summary: Check the method, units and conditions before writing your final answer.
 Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 7 — Indices and logarithms**7(a). Solve $(1/3)^{\sqrt{x}} = 81^{-x}$.****Step 1:** For real $\sqrt{x}$, $x \geq 0$. Rewrite with base 3: $3^{(-\sqrt{x})} = 3^{(-4x)}$, so $\sqrt{x} = 4x$.**Step 2:** Put $u = \sqrt{x} \geq 0$: $u = 4u^2$ gives $u(4u - 1) = 0$, so $u = 0$ or $1/4$.**Step 3:** Thus $x = u^2 = 0$ or $1/16$; both satisfy the original equation.**Final Answer: $x = 0$ or $x = 1/16$.**

Study Tip: Do not divide by $\sqrt{x}$ before checking $x = 0$, or you will lose a valid solution.

7(b). If $\log_{10}(x/5) = \log_{10}(2/x) + 1$, find x .**Step 1:** The logarithms require $x > 0$. Replace 1 with $\log_{10}10$.**Step 2:** Combine logs: $\log_{10}(x/5) = \log_{10}(20/x)$, so $x/5 = 20/x$.**Step 3:** Multiply by $5x$: $x^2 = 100$. The positive-domain condition keeps $x = 10$ and excludes -10 .**Final Answer: $x = 10$.**

Section Summary: Check the method, units and conditions before writing your final answer.
 Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 8 — Congruence and similarity

8(a). PX and QY are perpendicular to PQ, $PX = QY$, and X,A,Y are collinear. Prove triangles XPA and YQA congruent.

Step 1: $\angle XPA = \angle YQA = 90^\circ$, since both segments are perpendicular to PQ.

Step 2: $\angle XAP = \angle YAQ$ because they are vertically opposite angles.

Step 3: $PX = QY$ is given. Two corresponding angles and a corresponding side prove congruence.

Final Answer: $\triangle XPA \cong \triangle YQA$ by AAS (equivalently ASA after finding the third angles).

8(b). Triangles ABC and DEF are similar. $A = 60^\circ$, $B = 90^\circ$, $C = x$; $D = y$, $E = w$, $F = z$. Find x,y,z,w.

Step 1: In ABC, $x = 180 - 60 - 90 = 30^\circ$.

Step 2: From the stated order of similar triangles, $A \leftrightarrow D$, $B \leftrightarrow E$ and $C \leftrightarrow F$. Equal corresponding angles give $y = 60$, $w = 90$ and $z = 30$.

Final Answer: $x = 30^\circ$, $y = 60^\circ$, $z = 30^\circ$, $w = 90^\circ$.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 9 — Pythagoras and trigonometry

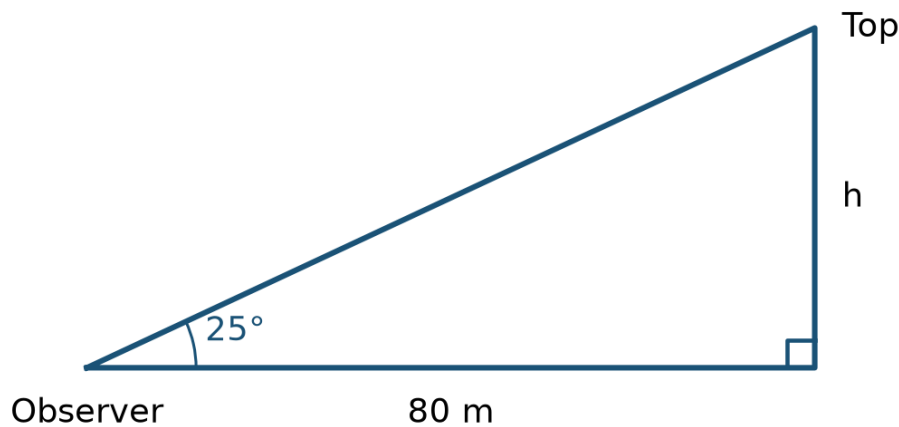
9(a). $PQ = 17$ cm, $QS = 15$ cm, $SR = 6$ cm, $PS \perp QR$ and $PR = x$. Find x.

Step 1: In right triangle PQS, $PS^2 = 17^2 - 15^2 = 289 - 225 = 64$, so $PS = 8$ cm.

Step 2: In right triangle PSR, $PR^2 = 8^2 + 6^2 = 100$, so $PR = 10$ cm.

Final Answer: $x = 10$ cm.

9(b). A point is 80 m from a vertical building. The angle of elevation to its top is 25° . Sketch and find the height to one decimal place.



Step 1: Draw a horizontal ground segment of 80 m, a vertical height h at the far end, and a sloping line of sight making 25° with the ground.

Step 2: $\tan 25^\circ = \text{opposite/adjacent} = h/80$; $h = 80 \tan 25^\circ$.

Step 3: Using a trigonometric table, $h \approx 80 \times 0.46631 = 37.3048$ m, which rounds to 37.3 m.

Final Answer: Height = 37.3 m.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Question 10 — Sets and statistics

10(a)(i). Of 30 students, 17 debate and 12 belong to both debate and Mathematics club. Find debate only.

Step 1: Debate total includes those in both activities.

Step 2: Debate only = $17 - 12 = 5$.

Final Answer: 5 students.

10(a)(ii). Every one of the 30 students participates in at least one activity. Find Mathematics club only.

Step 1: Divide the class into debate only, both and Mathematics only.

Step 2: Mathematics only = $30 - 5 - 12 = 13$; equivalently, $30 - 17 = 13$.

Final Answer: 13 students.

10(b)(i). Prepare a frequency table for ages:

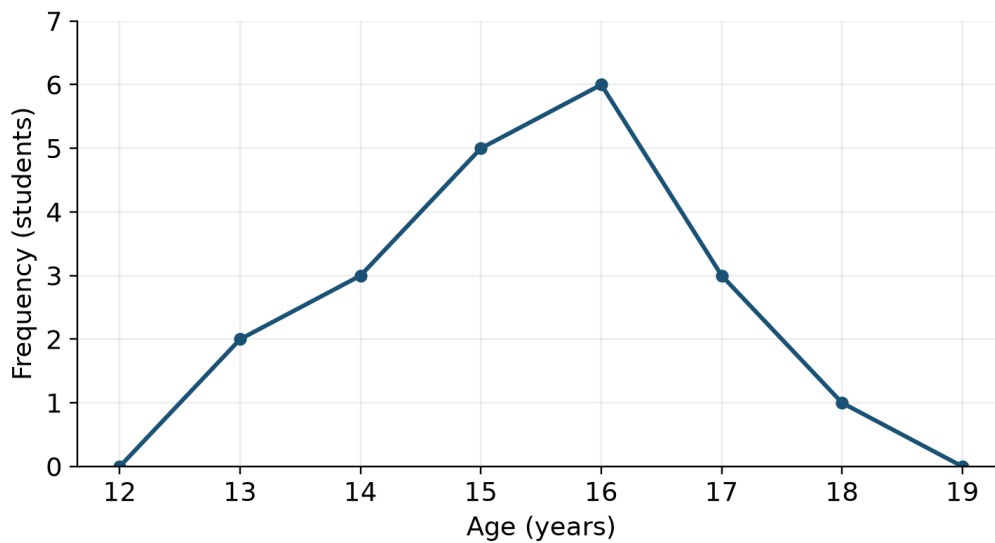
13,15,17,16,15,14,16,18,17,16,15,14,13,16,14,17,15,16,15,16.

Step 1: List distinct ages in ascending order and tally every occurrence.

Step 2: The counts are 2,3,5,6,3,1; check $2 + 3 + 5 + 6 + 3 + 1 = 20$.

Final Answer: Age (years): 13 | 14 | 15 | 16 | 17 | 18. Frequency: 2 | 3 | 5 | 6 | 3 | 1. Total: 20.

10(b)(ii). Draw a frequency polygon from the age-frequency table.



Step 1: Label the horizontal axis Age (years) and the vertical axis Frequency (students). Use equal scales.

Step 2: Plot one point for each age and its frequency, then join consecutive points with straight lines.

Step 3: Add zero-frequency end points at 12 and 19 to close the polygon.

Final Answer: Join (12,0), (13,2), (14,3), (15,5), (16,6), (17,3), (18,1), (19,0) with straight segments.

Section Summary: Check the method, units and conditions before writing your final answer. Substitute algebraic solutions back into the original equation and label all diagram measurements.

Source Checks

NECTA — FTNA 2021 Basic Mathematics: significant-figure convention and recurring-decimal notation